

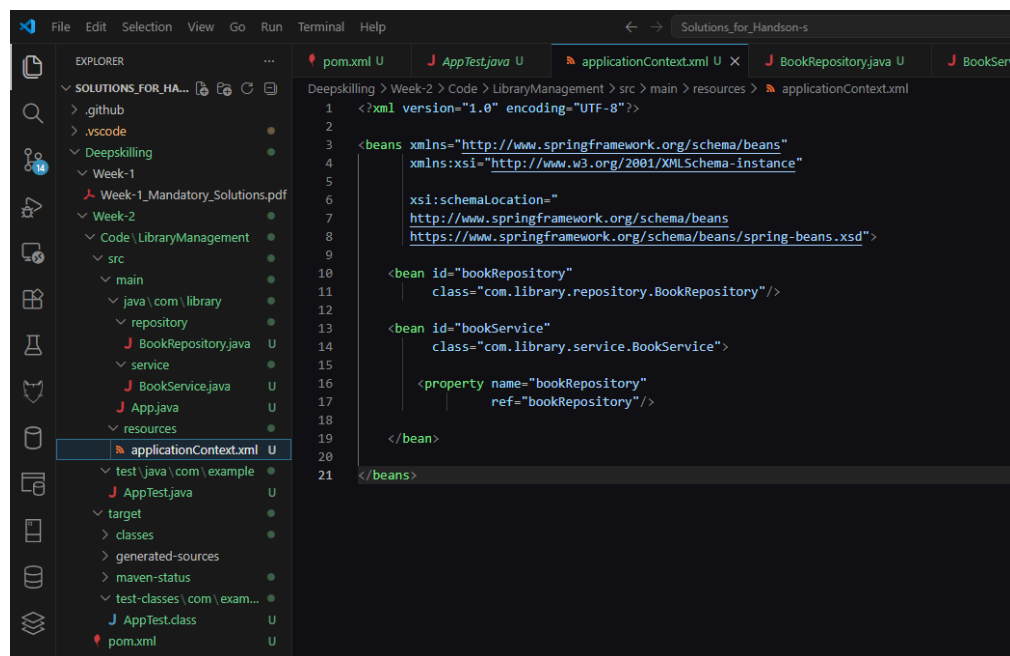
Spring Core_Maven

Exercise 1: Configuring a Basic Spring Application

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-2\Code\LibraryManagement> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.library:library_management >-----
[INFO] Building library_management 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ library_management ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ library_management ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-2\Code\LibraryManagement\target\classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.292 s
[INFO] Finished at: 2026-06-29T10:56:47+05:30
[INFO]
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-2\Code\LibraryManagement>
```

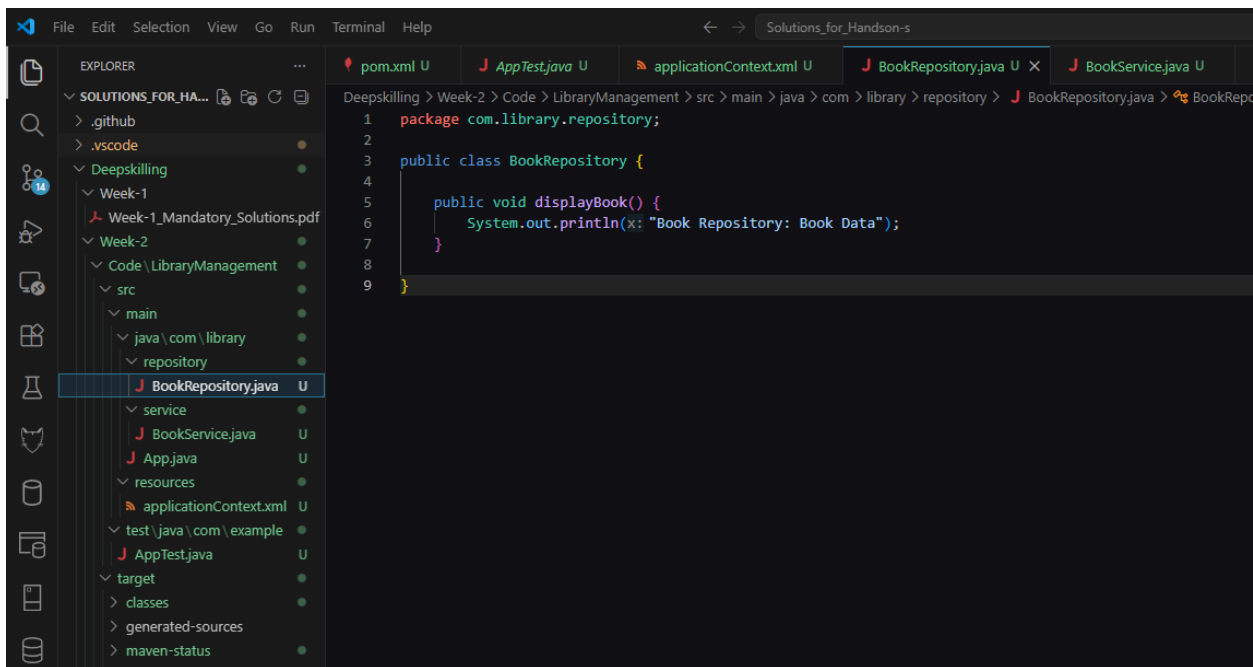
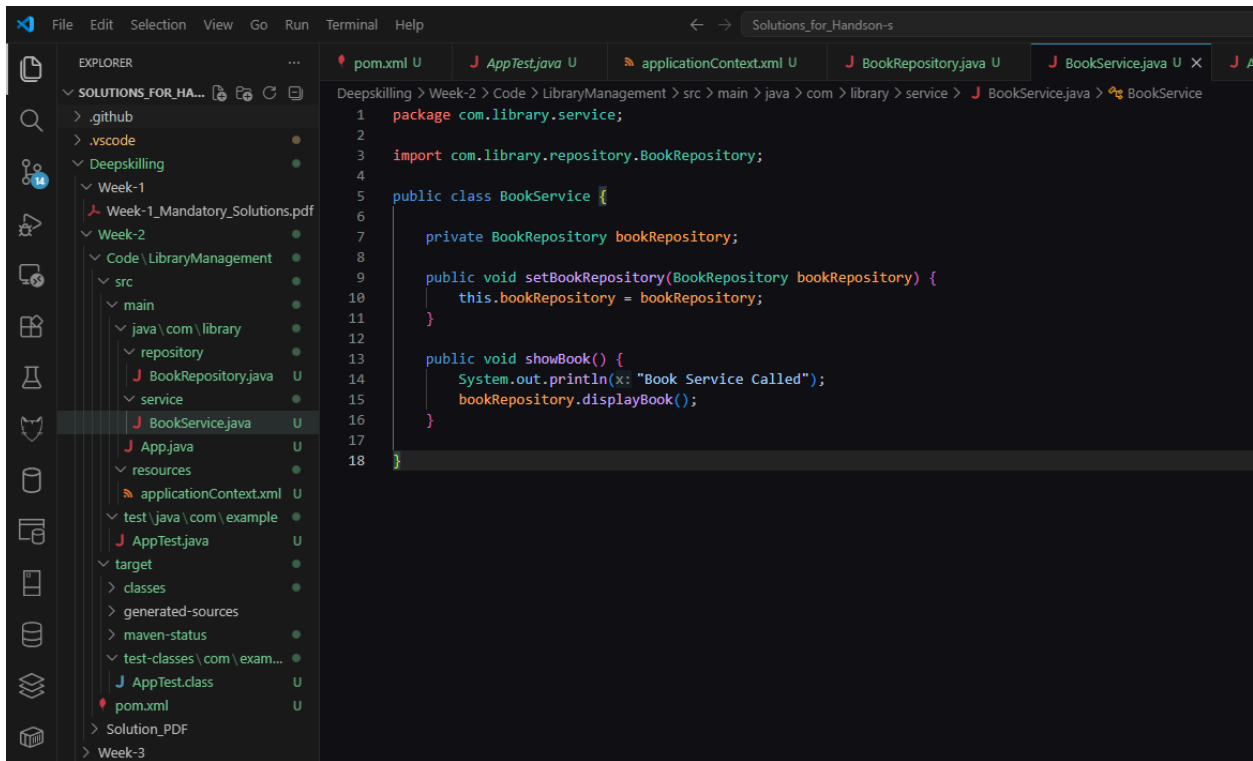
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  QUERY RESULTS
PS D:\Cogniz_folder\Solutions_for_Handson-s> & 'C:\Program Files\Java\jdk-17.0.10\bin\java.exe' '@C:\Users\lucal\AppData\Local\Temp\cp_54k11ywu3kwtc22jhtvzk2p.angfile' 'com.library.App'
Book Service Called
Book Repository: Book Data
PS D:\Cogniz_folder\Solutions_for_Handson-s>
```

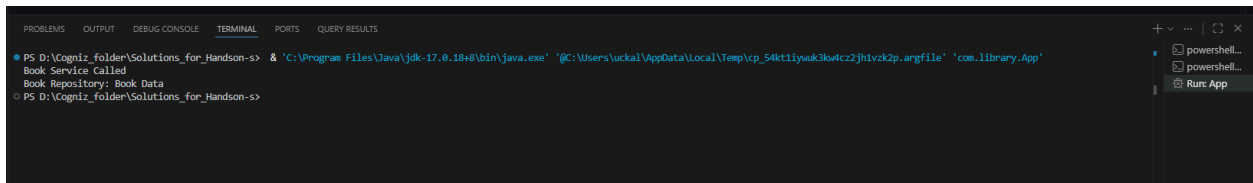
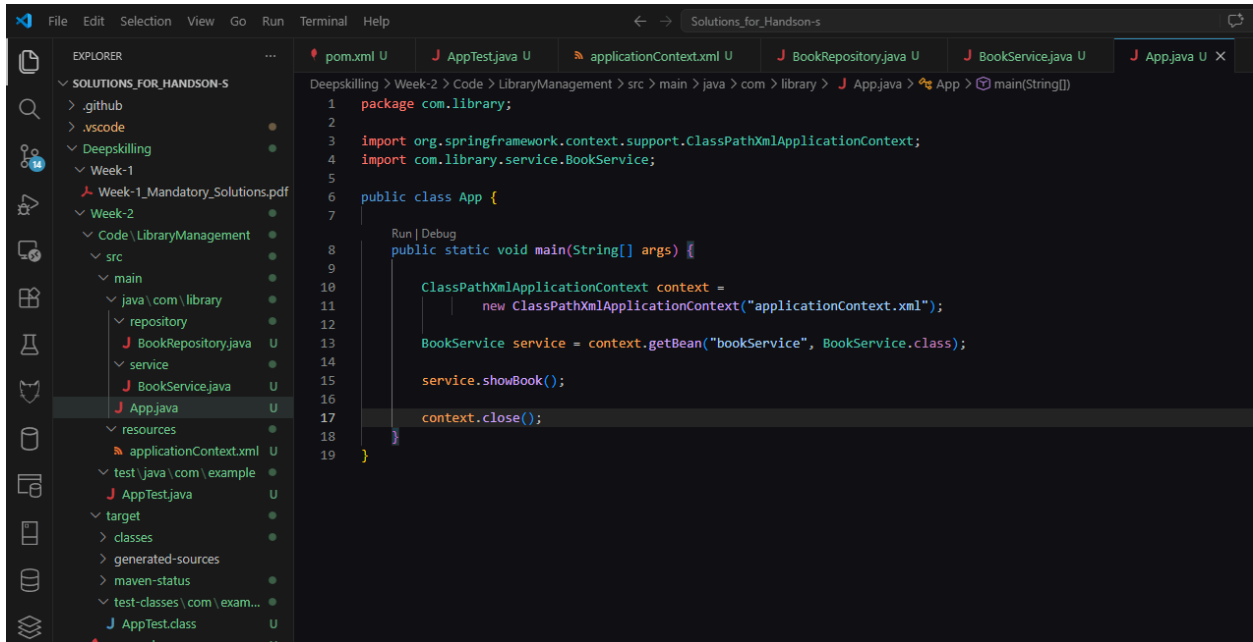
Exercise 2: Implementing Dependency Injection



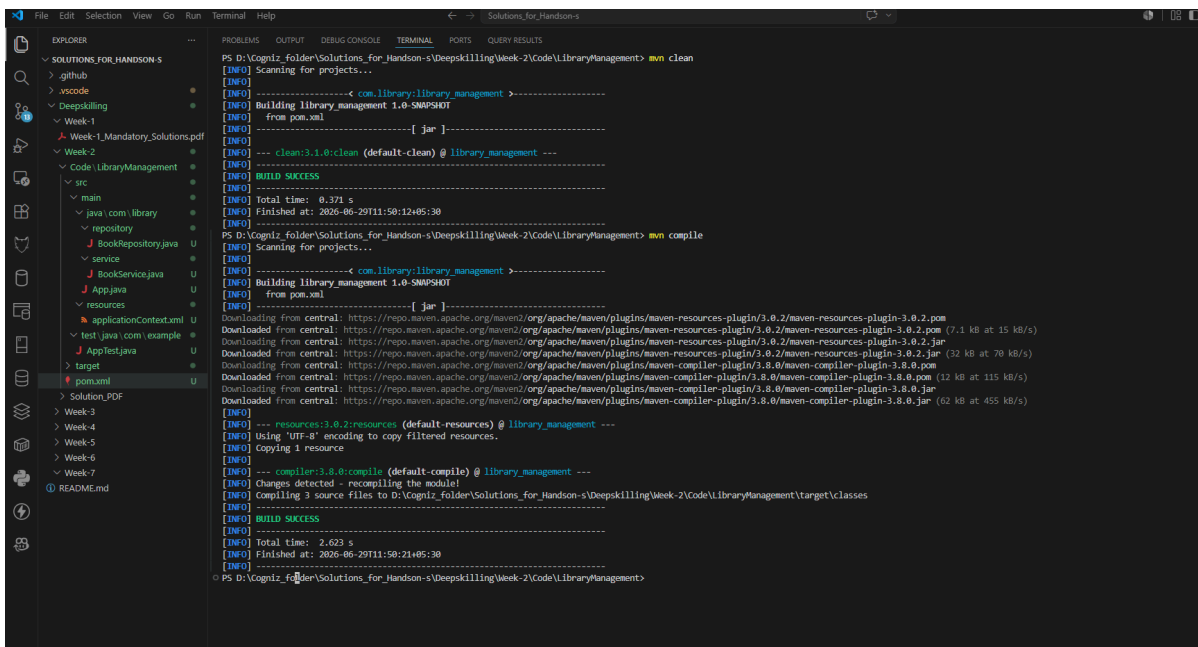
The screenshot shows the Visual Studio Code interface with the Explorer view on the left displaying the project structure. The main editor shows the `applicationContext.xml` file in the `src/main/resources` directory. The file contains an XML configuration for Spring beans.

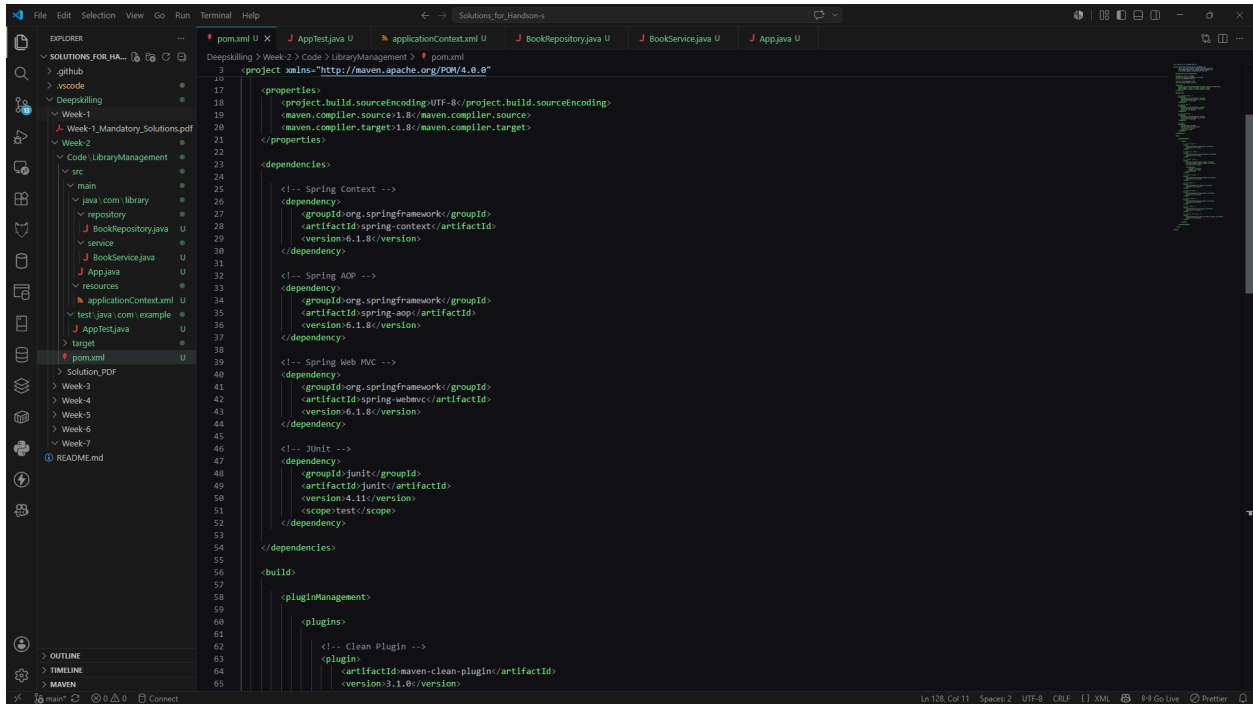
```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="bookRepository"
10          class="com.library.repository.BookRepository"/>
11
12     <bean id="bookService"
13          class="com.library.service.BookService">
14         <property name="bookRepository"
15                 ref="bookRepository"/>
16     </bean>
17
18 </beans>
```





Exercise 4: Creating and Configuring a Maven Project

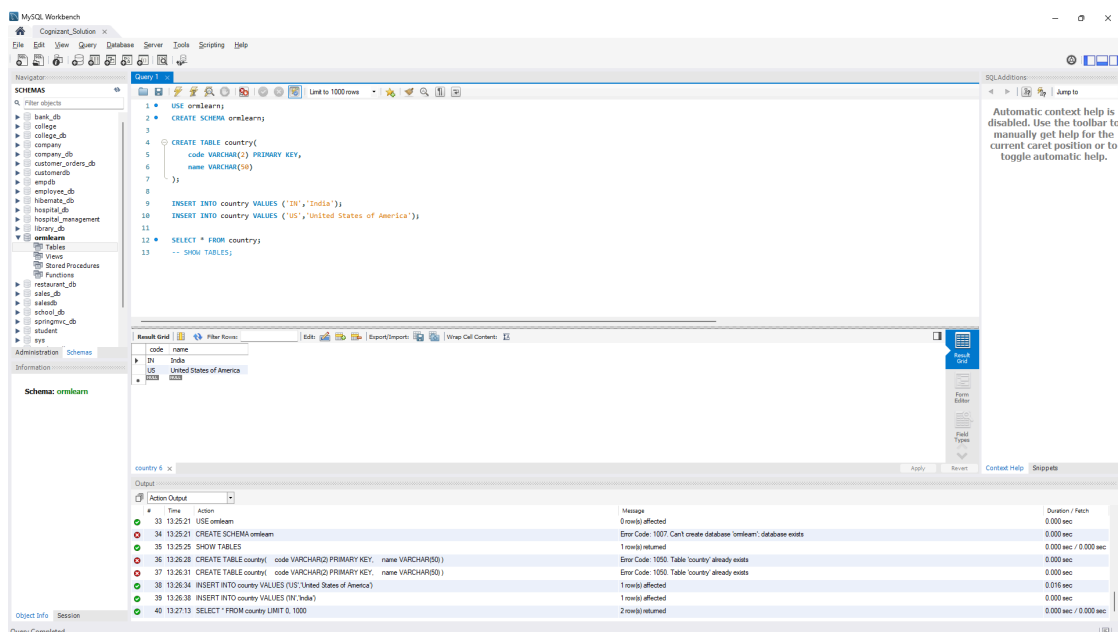


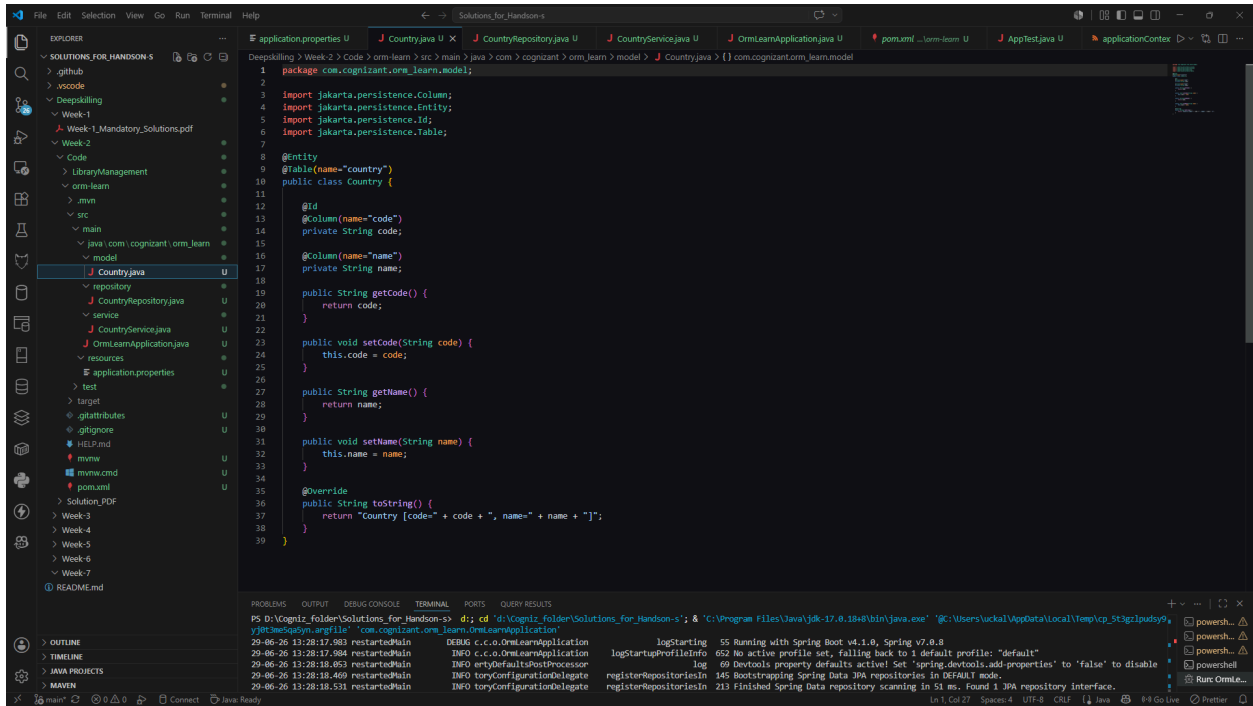
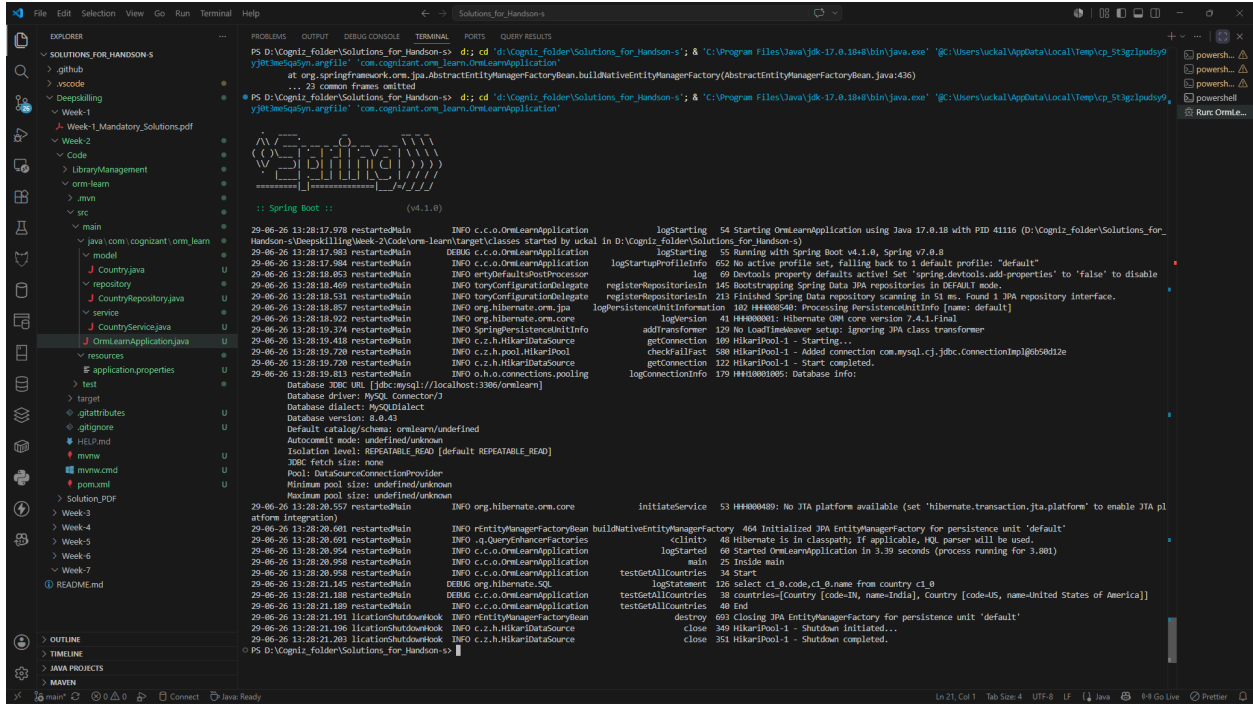


Spring Data JPA with Spring Boot, Hibernate

1. Spring-data-jpa-handson

Spring Data JPA - Quick Example





```
Help
Solutions_for_Handson-5

application.properties U X Country.java U CountryRepository.java U CountryService.java U OrmLearnApplication.java U pom.xml ...orm-learn U

Deepskilling > Week-2 > Code > orm-learn > src > main > resources > application.properties
1 # Spring Framework and application log
2 logging.level.org.springframework=info
3 logging.level.com.cognizant=debug
4
5 # Hibernate logs
6 logging.level.org.hibernate.SQL=trace
7 logging.level.org.hibernate.type.descriptor.sql=trace
8
9 # Log pattern
10 logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %Sp %-25.25logger{25} %25M %4L %m%n
11
12 # Database configuration
13 spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
14 spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
15 spring.datasource.username=root
16 spring.datasource.password=Ved@nt2295
17
18 # Hibernate configuration
19 spring.jpa.hibernate.ddl-auto=validate
```

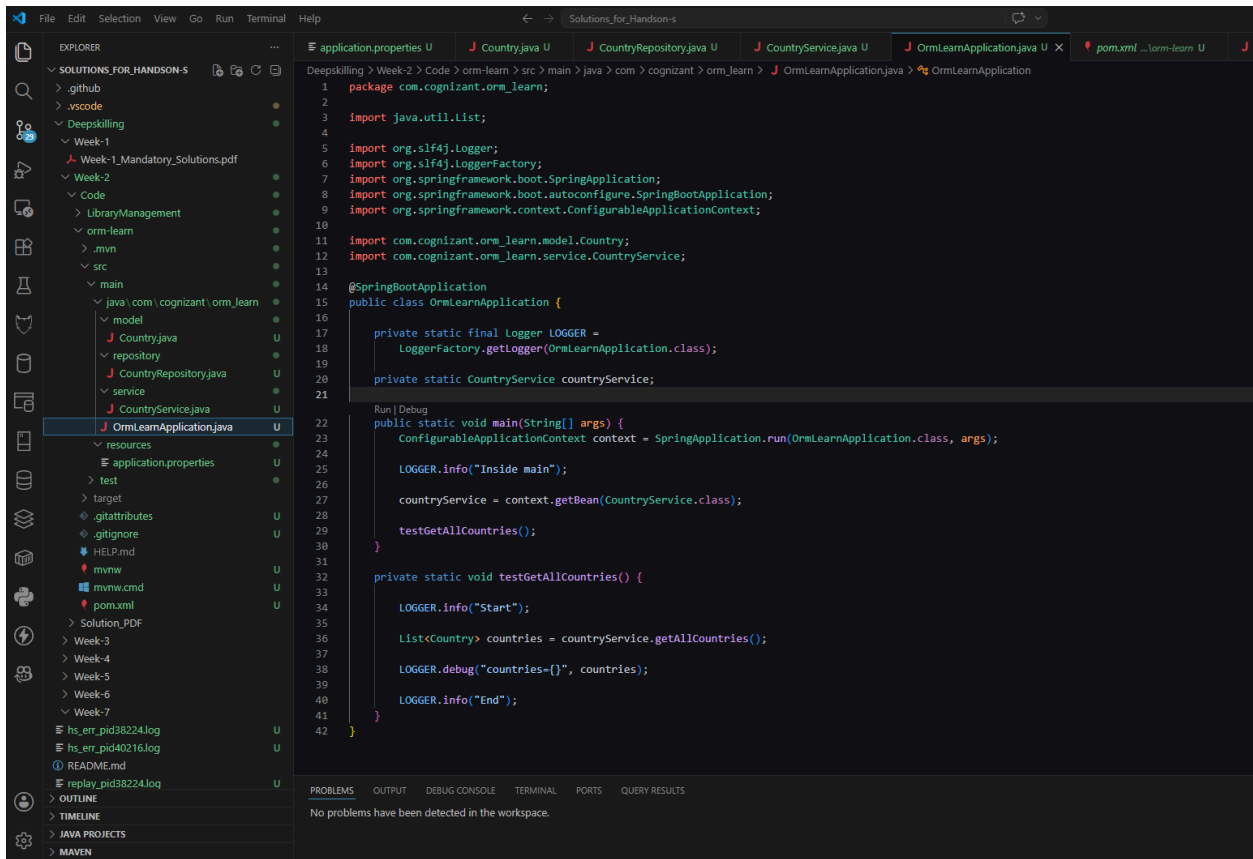
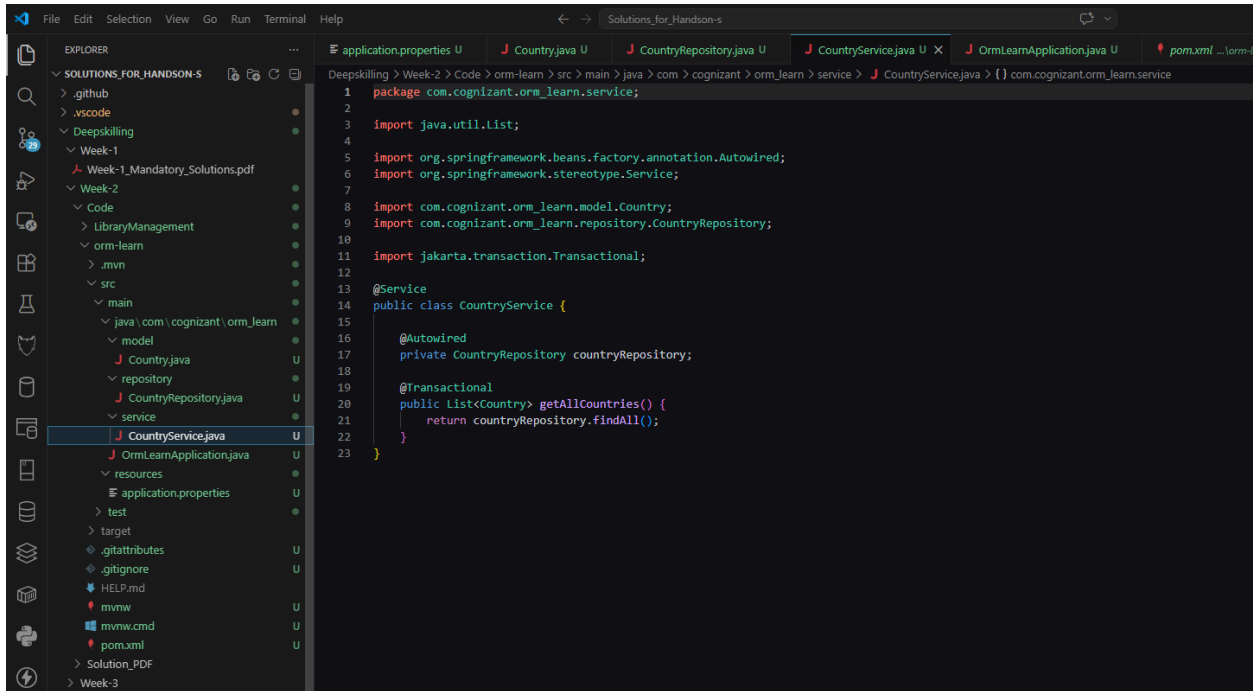
Difference between JPA, Hibernate and Spring Data JPA

```
File Edit Selection View Go Run Terminal Help
Solutions_for_Handson-5

EXPLORER
SOLUTIONS_FOR_HANDSON-5
  > .github
  > vscode
  > Deepskilling
  > Week-1
  > Week-1 Mandatory_Solutions.pdf
  > Week-2
  > Code
  > LibraryManagement
  > orm-learn
  > .mvn
  > src
  > main
  > java com cognizant orm_learn
  > model
  > Country.java
  > repository
  > CountryRepository.java
  > service
  > CountryService.java
  > OrmLearnApplication.java
  > resources
  > application.properties
  > test
  > target
  > .gitattributes
  > .gitignore
  > HELP.md
  > mvnw
  > mvnw.cmd
  > pom.xml
  > Solution_PDF
  > Week-3
  > Week-4
  > Week-5
  > Week-6
  > Week-7

Deepskilling > Week-2 > Code > orm-learn > src > main > java > com > cognizant > orm_learn > model > Country.java > Country
1 package com.cognizant.orm_learn.model;
2
3 import jakarta.persistence.Column;
4 import jakarta.persistence.Entity;
5 import jakarta.persistence.Id;
6 import jakarta.persistence.Table;
7
8 @Entity
9 @Table(name="country")
10 public class Country {
11
12     @Id
13     @Column(name="code")
14     private String code;
15
16     @Column(name="name")
17     private String name;
18
19     public String getCode() {
20         return code;
21     }
22
23     public void setCode(String code) {
24         this.code = code;
25     }
26
27     public String getName() {
28         return name;
29     }
30
31     public void setName(String name) {
32         this.name = name;
33     }
34
35     @Override
36     public String toString() {
37         return "Country [code=" + code + ", name=" + name + " ]";
38     }
39 }
```

```
Deepskilling > Week-2 > Code > orm-learn > src > main > java > com > cognizant > orm_learn > repository > CountryRepository.java
1 package com.cognizant.orm_learn.repository;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5
6 import com.cognizant.orm_learn.model.Country;
7
8 @Repository
9 public interface CountryRepository extends JpaRepository<Country,String> {
10
11 }
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS D:\Cogniz_folder\Solutions_for_Handson-sx & 'C:\Program Files\Java\jdk-17.0.18\bin\java.exe' '@C:\Users\uckal\AppData\Local\Temp\cp_5t3g2lpudsy9yj8t3me5qaSyn.argfile' 'com.cognizant.orm_learn.OrmLearnApplication'
```

```
=====|_|=====|_|_/_/_/_/
```

```
:: Spring Boot :: (v4.1.0)
```

```
29-06-26 14:57:21.573 restartedMain INFO c.c.o.OrmLearnApplication logStarting 54 Starting OrmLearnApplication using Java 17.0.18 with PID 58528 (D:\Cogniz_folder\Solutions_for_Handson-sx\Debug\Task-2\Code\orm_learn\target\classes started by uckal in D:\Cogniz_folder\Solutions_for_Handson-sx)
29-06-26 14:57:21.578 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 55 Running with Spring Boot v4.1.0, Spring v7.0.8
29-06-26 14:57:21.579 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 652 No active profile set, falling back to 1 default profile: "default"
29-06-26 14:57:21.652 restartedMain INFO ertyDefaultsPostProcessor log 69 Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
29-06-26 14:57:22.066 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 145 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
29-06-26 14:57:22.131 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 213 Finished Spring Data repository scanning in 54 ms. Found 1 JPA repository interface.
29-06-26 14:57:22.580 restartedMain INFO org.hibernate.orm.jpa logPersistenceUnitInformation 102 ###000540: Processing PersistenceUnitInfo [name: default]
29-06-26 14:57:22.590 restartedMain INFO org.hibernate.orm.core logVersion 41 ###000001: Hibernate ORM core version 7.4.1.Final
29-06-26 14:57:23.127 restartedMain INFO SpringPersistenceUnitInfo addTransformer 129 No LoadTimeWeaver setup: ignoring JPA class transformer
29-06-26 14:57:23.164 restartedMain INFO c.z.h.HikariDataSource getConnection 109 HikariPool-1 - Starting...
29-06-26 14:57:23.676 restartedMain INFO c.z.h.pool.HikariPool checkFailFast 580 HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@6b58d12e
29-06-26 14:57:23.679 restartedMain INFO c.z.h.HikariDataSource getConnection 122 HikariPool-1 - Start completed.
29-06-26 14:57:23.824 restartedMain INFO o.h.o.connections.pooling logConnectionInfo 179 ###0001005: Database Info:
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.43
Default catalog/schema: ormLearn/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
29-06-26 14:57:24.826 restartedMain INFO org.hibernate.orm.core initiateService 53 ###000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
29-06-26 14:57:24.912 restartedMain INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for persistence unit 'default'
29-06-26 14:57:25.014 restartedMain INFO .q.QueryEnhancerFactories <init> 48 Hibernate is in classpath; if applicable, HQL parser will be used.
29-06-26 14:57:25.338 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearnApplication in 4.209 seconds (process running for 4.747)
29-06-26 14:57:25.342 restartedMain INFO c.c.o.OrmLearnApplication main 25 Inside main
29-06-26 14:57:25.343 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 34 Start
29-06-26 14:57:25.638 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1.0.code,c1.0.name from country c1.0
29-06-26 14:57:25.702 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 38 countries=[country [code=IN, name=India], Country [code=US, name=United States of America]]
29-06-26 14:57:25.703 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 40 End
29-06-26 14:57:25.712 licationShutdownHook INFO rEntityManagerFactoryBean destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
29-06-26 14:57:25.718 licationShutdownHook INFO c.z.h.HikariDataSource close 349 HikariPool-1 - Shutdown initiated...
29-06-26 14:57:25.736 licationShutdownHook INFO c.z.h.HikariDataSource close 351 HikariPool-1 - Shutdown completed.
```

```
PS D:\Cogniz_folder\Solutions_for_Handson-sx
```